DRAFT – REQUIRES STUDENT CONFIRMATION. This text describes the proposed division of work. The student must check every item, change it so that it describes what they actually did (including, where relevant, which parts were prepared with the help of an AI assistant), fill in every red placeholder, and then switch off this banner (`workdone/status.tex`) before submission.

I worked on the following parts in this project:

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1. **The `while` loop and the variable s.** I proved Lemma 1 (line 7 runs $\lambda = \lfloor \log_2 n \rfloor + 1$ times and $\lambda \leq n$) and Lemma 2 (what s counts), and made the per-iteration cost table of Section 3.3 together with the pair-counting identity $\sum_{\mathtt{b}}(s_i + 1) = n_b + P$.

2. **Theoretical analysis of group (3).** I derived $\tau_3 = n + P$, $B_3(n) = n$, and $W_3(n) = n + \lfloor n^2/4 \rfloor$ with worst input $\mathtt{a}^{\lfloor n/2 \rfloor}\mathtt{b}^{\lceil n/2 \rceil}$, using $P \leq n_a n_b$ and the maximisation of $x(n-x)$; and $A_3(n) = \frac{9n}{5} - 4 + 4(\frac{4}{5})^n$ with its $\Theta(n)$ bounds (Section 6).

3. **Theoretical analysis of group (4).** I proved the per-iteration lower bound $\tau_{4,i} \geq \min(\lambda, n-i)$, which gives $B_4(n) = n\lambda - \lambda(\lambda-1)/2$ with the input $\mathtt{a}^{n-\lambda}\mathtt{c}^{\lambda}$; the upper bound $\tau_4 \leq f(x) = \lambda x + n(n-x)(x+1)$ and its maximisation through $f(x+1) - f(x) = \lambda + n(n-2x-2)$, which gives $W_4(n)$ and the input $\mathtt{a}^{\lfloor n/2 \rfloor}\mathtt{b}^{\lceil n/2 \rceil}$; and $A_4(n)$ with the $\Theta(n \log n)$, $\Theta(n^3)$ and $\Theta(n^2)$ bounds (Section 7). I filled Table 1.

4. **Independent verification of the formulas.** I wrote the separate pseudocode interpreter (loops expanded into `while` form, counts per line), the exhaustive check over all 3^n inputs for $n \leq 10$, the dynamic programs over (i, s) for the exact optimum ($n \leq 120$) and the exact expectation, and the `sympy` check (Section 8.1).

5. **Exact count verification.** I ran the best/worst inputs of every group for $n \in \{1, 5, 10, 20, 40, 100\}$ (Table 3) and 1000 random inputs for $n \in \{10, 40, 100\}$ (Table 4), computed the standard errors and wrote the comment on the differences (Section 11).

6. **Graphs and comparison.** I made the five figures from the result files and wrote Section 12: the comments on all twelve analyses, the comparison of the average and worst case, and the discussion of small inputs and noise.

7. **Tests.** I wrote the exhaustive tests against the interpreter and the tests of the formulas and of the random input distribution.

8. **Cross-checking.** [STUDENT TO FILL IN: which parts of Mert's work (e.g. the derivation of $\mathrm{E}[S_i]$ and $A_2(n)$) you checked, and how]. I read the final report.